THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4	$AE = k \times AD$
2	**(c)** 5	Perimeter ratio = side ratio
3	**(d)** 6	$h^2 = pq$
4	**(d)** 36	Similar triangles
5	**(d)** 100	Half-diagonals are legs
6	**(a)** 13	Right triangle with legs d and $a - b$
7	**(a)** 9	$AE = k \times AD$
8	**(c)** 7	Perimeter ratio = side ratio
9	**(c)** 6	$h^2 = pq$
10	**(d)** 64	Similar triangles

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{3 \times 10}{5} = 6$ cm (1); $EC = 10 - 6 = 4$ cm (1).	1+1+1
12	Ratio of sides = ratio of perimeters = $1 : 2$ (1); side = 8 cm (1); ratio of areas = $1 : 4$ (1).	1+1+1
13	Midpoint theorem: $DE \parallel BC$, $DE = \frac{1}{2}BC = 4$ cm (2); areas in ratio $(1/2)^2 = 1 : 4$ (1).	2+1
14	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{4 \times 20}{10} = 8$ cm (1); $EC = 20 - 8 = 12$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{4 \times 18}{9} = 8$ cm (1); $EC = 18 - 8 = 10$ cm (1).	1+1+1
16	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 2$ (1); $QR = 8$ cm (1); $RP = 10$ cm (1).	1+1+1
17	Altitude^2 = product of the segments (1); $h^2 = 4 \times 9 = 36$ (1); $h = 6$ cm (1).	1+1+1
	OR: $\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{2 \times 27}{9} = 6$ cm (1); $EC = 27 - 6 = 21$ cm (1).	1+1+1
18	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 3$ (1); $QR = 18$ cm (1); $RP = 21$ cm (1).	1+1+1
	OR: Altitude^2 = product of the segments (1); $h^2 = 2 \times 8 = 16$ (1); $h = 4$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw a horizontal line from the top of the shorter pole: right triangle with legs 9 and 12 (2); $x^2 = 81 + 144$ (1); $x = 15$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 6 and 8 (1); $s^2 = 6^2 + 8^2 = 100$ (1); side $= \sqrt{100}$ cm (1).	1+1+1
20	Draw a horizontal line from the top of the shorter pole: right triangle with legs 16 and 12 (2); $x^2 = 256 + 144$ (1); $x = 20$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 5 and 12 (1); $s^2 = 5^2 + 12^2 = 169$ (1); side $= \sqrt{169}$ cm (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).
- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 169	Half-diagonals are legs
2	**(a)** 10	Right triangle with legs d and $a-b$
3	**(d)** 12	$AE = k \times AD$
4	**(b)** 9	Perimeter ratio = side ratio
5	**(a)** 6	$h^2 = pq$
6	**(a)** 90	Similar triangles
7	**(c)** 225	Half-diagonals are legs
8	**(a)** 25	Right triangle with legs d and $a-b$
9	**(d)** 8	$AE = k \times AD$
10	**(d)** 6	Perimeter ratio = side ratio

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ratio of sides = ratio of perimeters $= 1:3$ (1); side $= 15$ cm (1); ratio of areas $= 1:9$ (1).	1+1+1
12	Midpoint theorem: $DE \parallel BC$, $DE = \frac{1}{2}BC = 6$ cm (2); areas in ratio $(1/2)^2 = 1:4$ (1).	2+1
13	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{2 \times 15}{5} = 6$ cm (1); $EC = 15 - 6 = 9$ cm (1).	1+1+1
14	Ratio of sides = ratio of perimeters $= 1:4$ (1); side $= 12$ cm (1); ratio of areas $= 1:16$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{6 \times 18}{9} = 12$ cm (1); $EC = 18 - 12 = 6$ cm (1).	1+1+1
16	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 2$ (1); $QR = 14$ cm (1); $RP = 16$ cm (1).	1+1+1
17	Altitude^2 = product of the segments (1); $h^2 = 3 \times 27 = 81$ (1); $h = 9$ cm (1).	1+1+1
	OR: $\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{5 \times 26}{13} = 10$ cm (1); $EC = 26 - 10 = 16$ cm (1).	1+1+1
18	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 4$ (1); $QR = 32$ cm (1); $RP = 40$ cm (1).	1+1+1
	OR: Altitude^2 = product of the segments (1); $h^2 = 5 \times 20 = 100$ (1); $h = 10$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw a horizontal line from the top of the shorter pole: right triangle with legs 5 and 12 (2); $x^2 = 25 + 144$ (1); $x = 13$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 9 and 12 (1); $s^2 = 9^2 + 12^2 = 225$ (1); side $= \sqrt{225}$ cm (1).	1+1+1
20	Draw a horizontal line from the top of the shorter pole: right triangle with legs 12 and 16 (2); $x^2 = 144 + 256$ (1); $x = 20$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 8 and 15 (1); $s^2 = 8^2 + 15^2 = 289$ (1); side $= \sqrt{289}$ cm (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).
- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10	$h^2 = pq$
2	**(b)** 45	Similar triangles
3	**(d)** 289	Half-diagonals are legs
4	**(b)** 17	Right triangle with legs d and $a - b$
5	**(d)** 27	$AE = k \times AD$
6	**(a)** 8	Perimeter ratio = side ratio
7	**(c)** 12	$h^2 = pq$
8	**(b)** 60	Similar triangles
9	**(b)** 625	Half-diagonals are legs
10	**(c)** 29	Right triangle with legs d and $a - b$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Midpoint theorem: $DE \parallel BC$, $DE = \frac{1}{2}BC = 9$ cm (2); areas in ratio $(1/2)^2 = 1 : 4$ (1).	2+1
12	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{5 \times 20}{10} = 10$ cm (1); $EC = 20 - 10 = 10$ cm (1).	1+1+1
13	Ratio of sides = ratio of perimeters $= 1 : 5$ (1); side $= 10$ cm (1); ratio of areas $= 1 : 25$ (1).	1+1+1
14	Midpoint theorem: $DE \parallel BC$, $DE = \frac{1}{2}BC = 11$ cm (2); areas in ratio $(1/2)^2 = 1 : 4$ (1).	2+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{3 \times 39}{13} = 9$ cm (1); $EC = 39 - 9 = 30$ cm (1).	1+1+1
16	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 3$ (1); $QR = 27$ cm (1); $RP = 36$ cm (1).	1+1+1
17	Altitude^2 = product of the segments (1); $h^2 = 8 \times 18 = 144$ (1); $h = 12$ cm (1).	1+1+1
	OR: $\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{7 \times 22}{11} = 14$ cm (1); $EC = 22 - 14 = 8$ cm (1).	1+1+1
18	$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 2$ (1); $QR = 20$ cm (1); $RP = 22$ cm (1).	1+1+1
	OR: Altitude^2 = product of the segments (1); $h^2 = 9 \times 16 = 144$ (1); $h = 12$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw a horizontal line from the top of the shorter pole: right triangle with legs 35 and 12 (2); $x^2 = 1225 + 144$ (1); $x = 37$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 12 and 16 (1); $s^2 = 12^2 + 16^2 = 400$ (1); side $= \sqrt{400}$ cm (1).	1+1+1
20	Draw a horizontal line from the top of the shorter pole: right triangle with legs 15 and 8 (2); $x^2 = 225 + 64$ (1); $x = 17$ m (2).	2+1+2
	OR: Diagonals bisect at right angles: half-diagonals 7 and 24 (1); $s^2 = 7^2 + 24^2 = 625$ (1); side $= \sqrt{625}$ cm (1).	1+1+1

Mistakes to avoid:

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